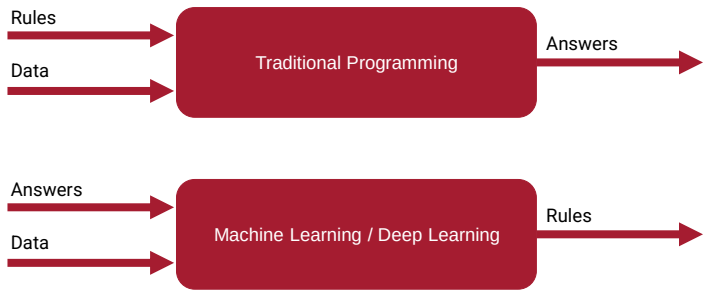
AI Edge (Computing)

2. Getting started with Deep Learning (DL)

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1

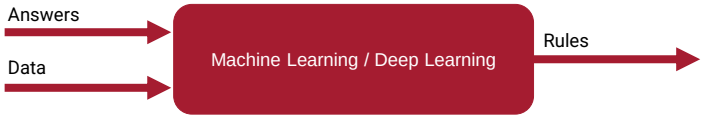
Traditional programming vs ML/DL



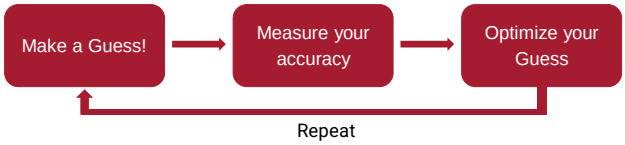
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2

Figuring out the rules



Determining the rules:



3

Rules and loss

Example:

(input) X: -1, 0, 1, 2, 3, 4

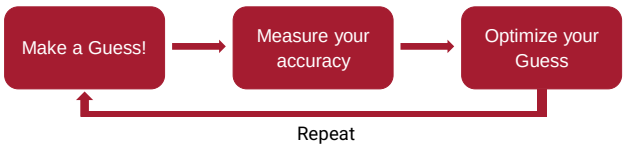
(output) Y: -3, -1, 1, 3, 5, 7

What is the relationship? What is the rule to go from X to Y?

Try: Y = 3X-1

Yguess : -4, -1, 2, 5, 8, 11

Fail! Retry!



4

Rules and loss

How do we do this with numbers?

Determining loss

(input) X = -1, 0, 1, 2, 3, 4

(output) Y = -3, -1, 1, 3, 5, 7

With Y = 3X-1 → Yguess = -4, -1, 2, 5, 8, 11

Error = Yguess – Y = -1, 0, 1, 2, 3, 4

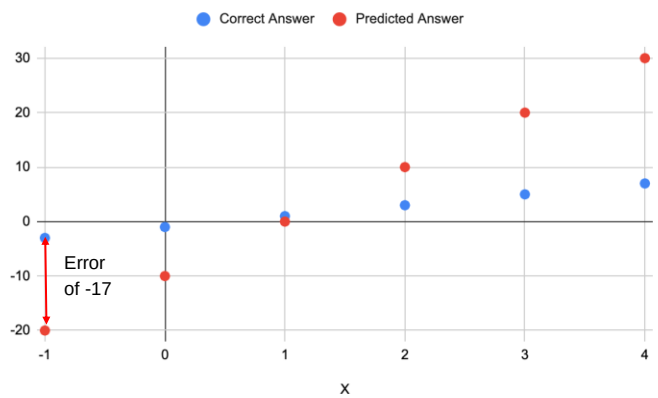
Squared errors = 1, 0, 1, 4, 9, 16

Mean squared error (MSE) = 1/6*(1+0+1+4+9+16) = 5,1667

5

Rules and loss

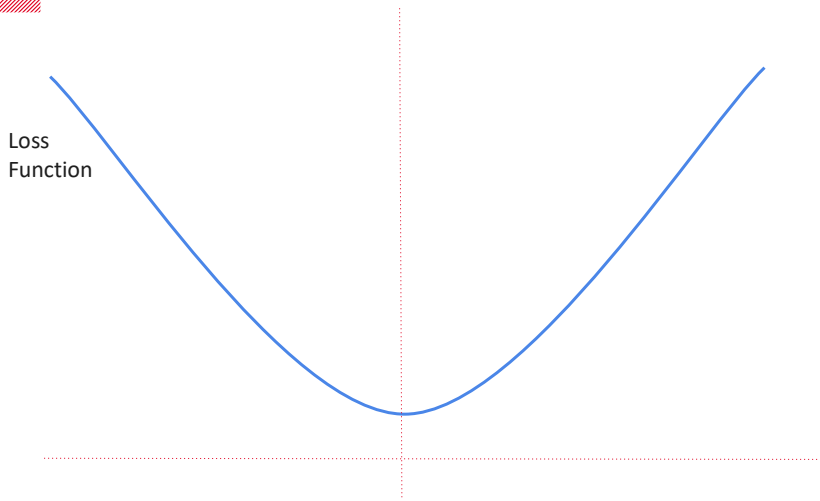
What if we guessed: Y = 10X-10 ?



6

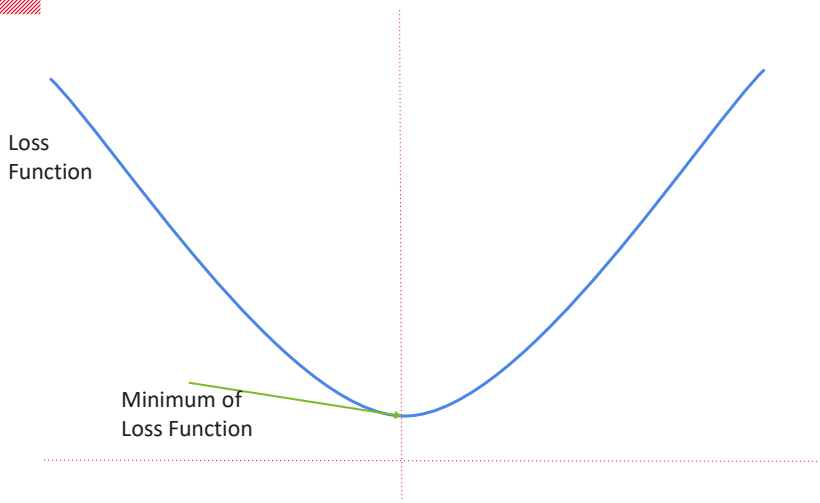
Learning rate and loss

Learning rate is the factor which we multiply with the gradient to adjust the biasses and the weights



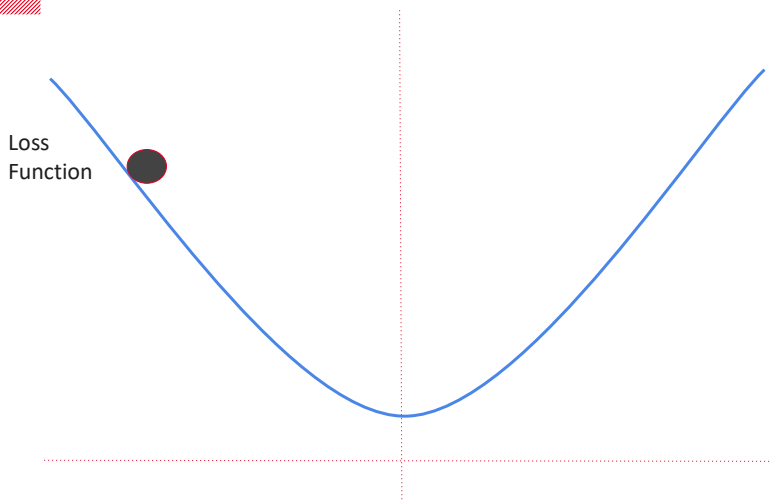
7

Learning rate and loss



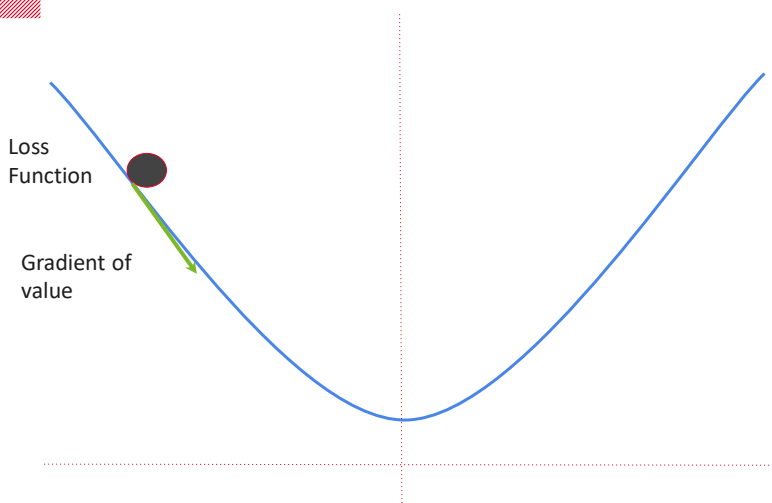
8

Learning rate and loss



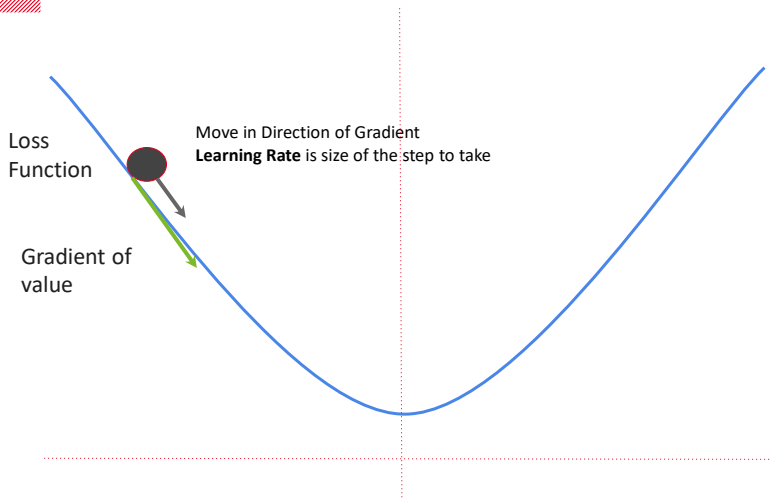
9

Learning rate and loss



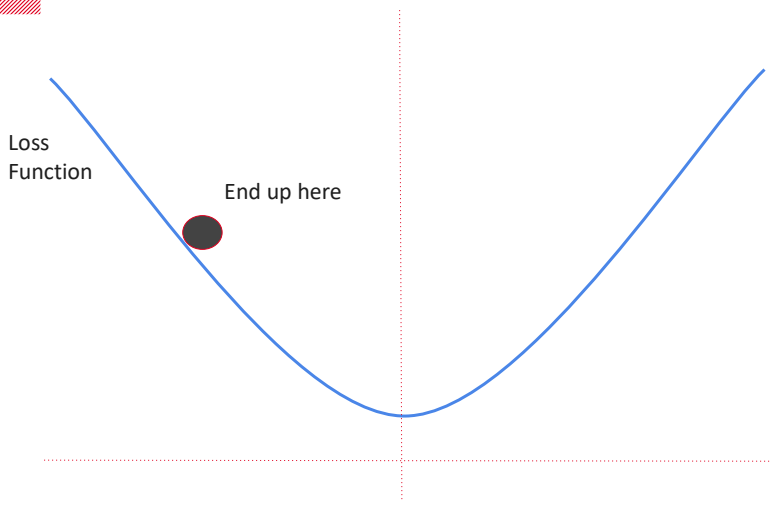
10

Learning rate and loss



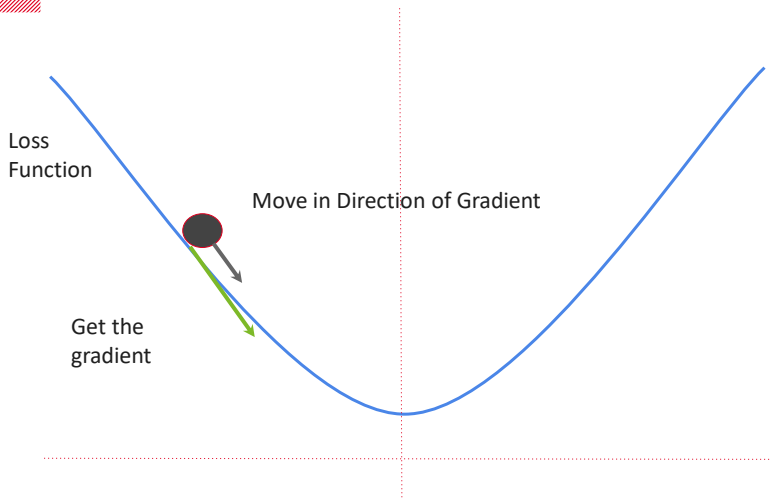
11

Learning rate and loss



12

Learning rate and loss



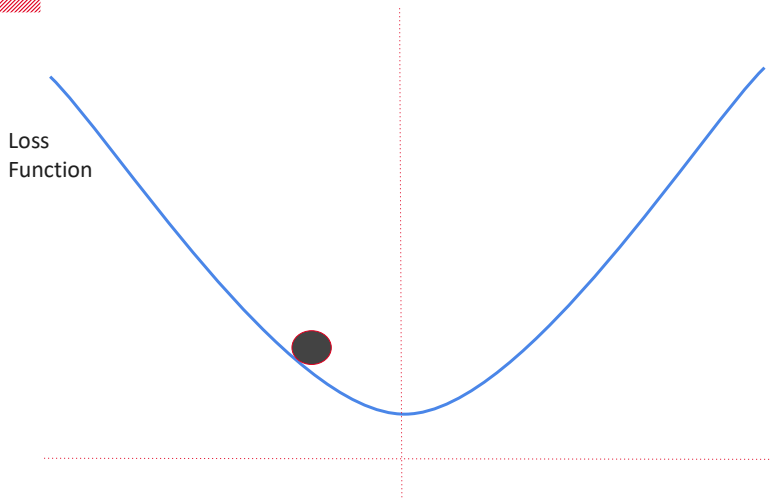
13

Learning rate and loss



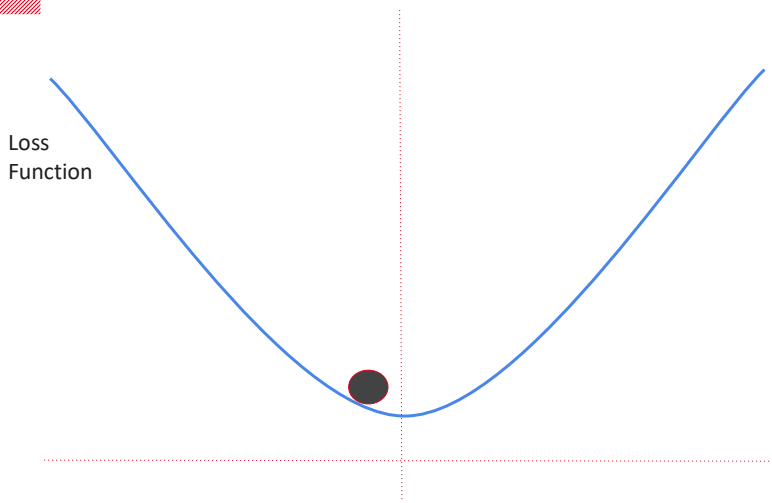
14

Learning rate and loss



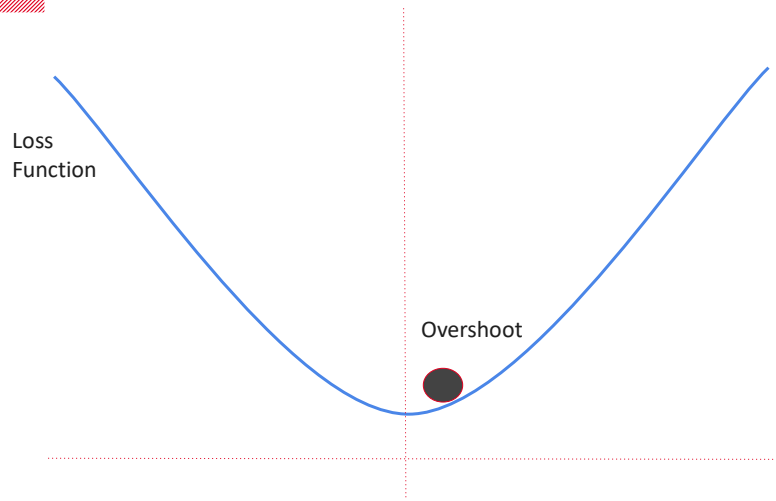
15

Learning rate and loss



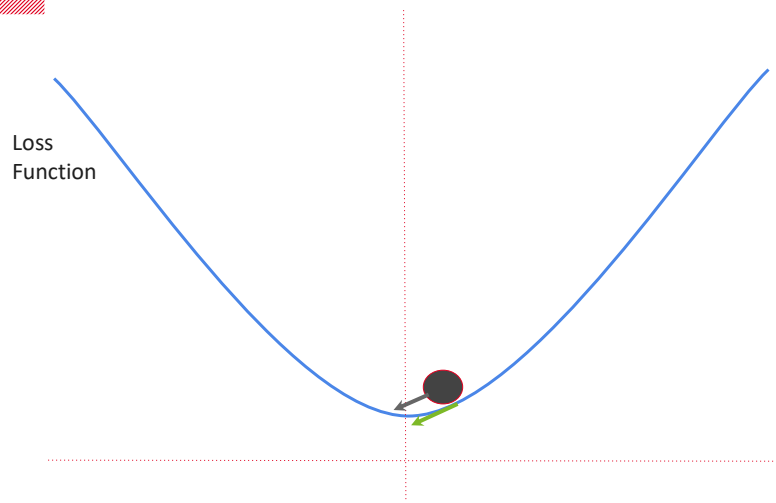
16

Learning rate and loss



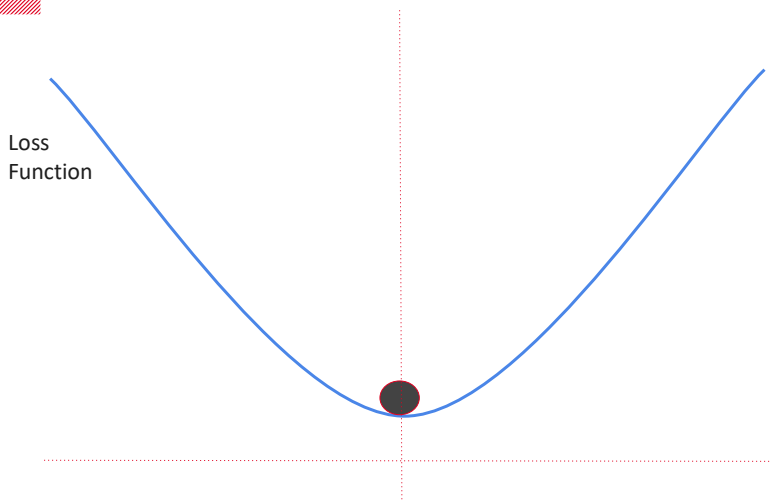
17

Learning rate and loss



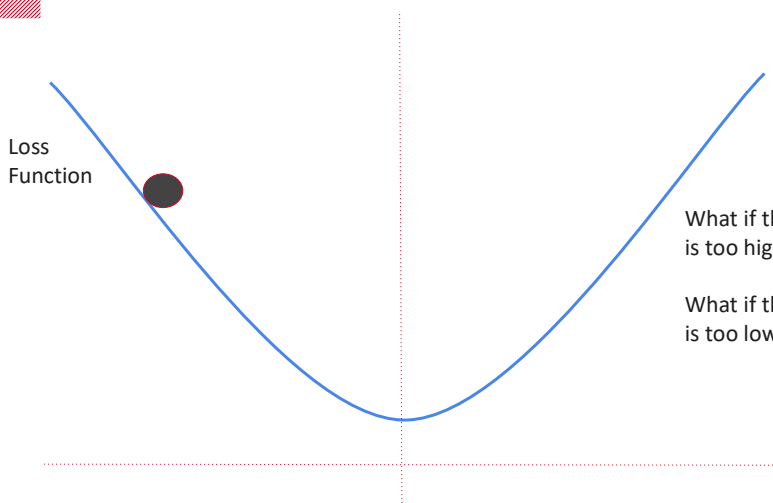
18

Learning rate and loss



19

Learning rate and loss

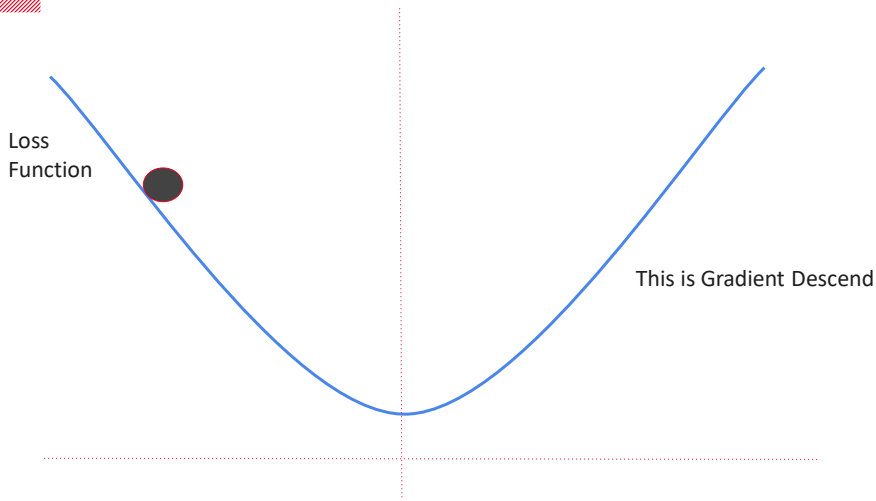


What if the learning rate is too high?

What if the learning rate is too low?

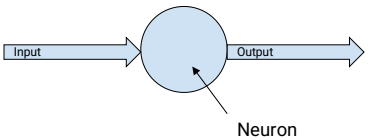
20

Learning rate and loss



21

Example in Python




```
model = keras.Sequential([keras.layers.Dense(units=1, input_shape=[1])])
model.compile(optimizer='sgd', loss='mean_squared_error')

xs = np.array([-1.0, 0.0, 1.0, 2.0, 3.0, 4.0], dtype=float)
ys = np.array([-3.0, -1.0, 1.0, 3.0, 5.0, 7.0], dtype=float)

model.fit(xs, ys, epochs=500)

print(model.predict([10.0]))
```


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Some questions

In machine learning, programmers write explicit rules to match answers to data.

a) True

b) False

23

Some questions

Which of the following are true about learning rates? Select all that apply.

a) If it is too large, we may never reach the minimum

b) If it is too small, we will never reach the minimum

c) If it is too small, it may take a long time to reach the minimum

d) If it just right, the loss is guaranteed to be 0

25

Some questions

Models can be made to generalize patterns in data by minimizing the loss function.

a) True

b) False

27

Some questions

A dense neural network can be used to compute linear regression.

a) True

b) False

29

Some questions

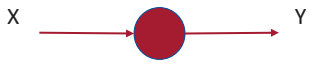
Neural network parameters must be manually computed by the programmer.

a) True

b) False

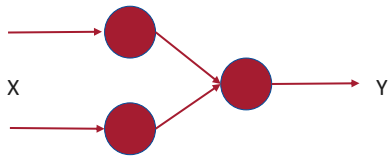
31

From regression to classification



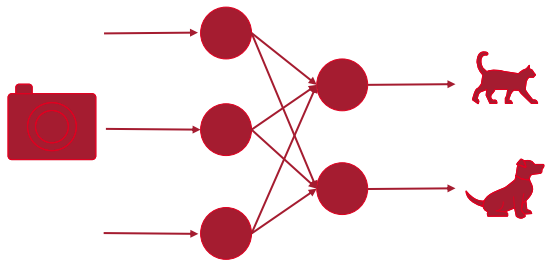
33

From regression to classification



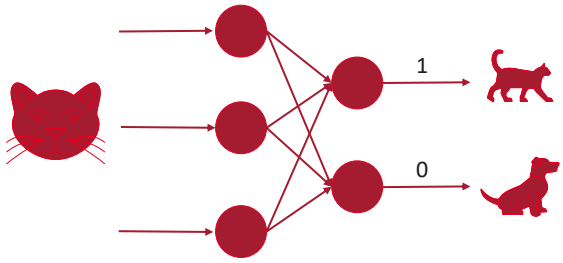
34

From regression to classification



35

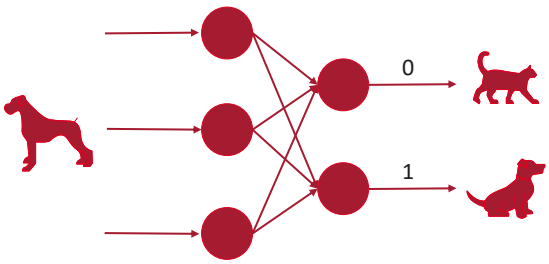
From regression to classification



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From regression to classification



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37

From regression to classification

Data	Label
	[1, 0]
	[0, 1]

38

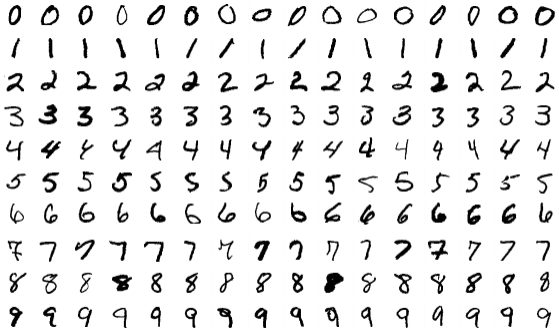
From regression to classification

0	[1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
1	[0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
2	[0, 0, 1, 0, 0, 0, 0, 0, 0, 0]
3	[0, 0, 0, 1, 0, 0, 0, 0, 0, 0]
4	[0, 0, 0, 0, 1, 0, 0, 0, 0, 0]
5	[0, 0, 0, 0, 0, 1, 0, 0, 0, 0]
6	[0, 0, 0, 0, 0, 0, 1, 0, 0, 0]
7	[0, 0, 0, 0, 0, 0, 0, 1, 0, 0]
8	[0, 0, 0, 0, 0, 0, 0, 0, 1, 0]
9	[0, 0, 0, 0, 0, 0, 0, 0, 0, 1]

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Example MNIST

Modified National Institute of Standards and Technology database

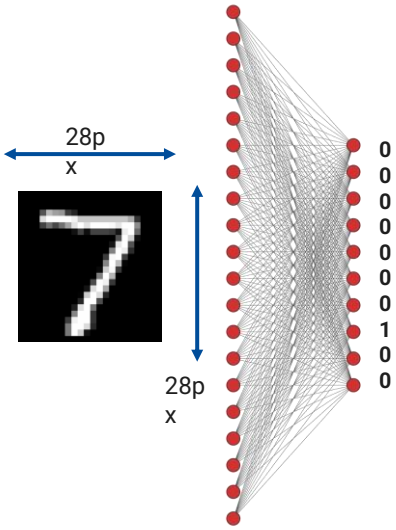


60,000 Labelled Training Examples
10.000 Labelled Validation Examples

40

Example MNIST

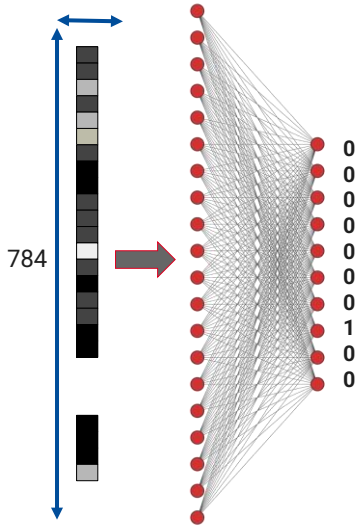
Modified National Institute of Standards and Technology database



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Example MNIST

Modified National Institute of Standards and Technology database



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Example MNIST

Modified National Institute of Standards and Technology database

```
import tensorflow as tf

data = tf.keras.datasets.mnist
(training_images, training_labels), (val_images, val_labels) = data.load_data()

training_images  = training_images / 255.0
val_images = val_images / 255.0

model = tf.keras.models.Sequential(
    [tf.keras.layers.Flatten(input_shape=(28,28)),
     tf.keras.layers.Dense(20, activation=tf.nn.relu),
     tf.keras.layers.Dense(10, activation=tf.nn.softmax)])
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(training_images, training_labels, epochs=20)
```


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Example MNIST

Modified National Institute of Standards and Technology database

```
classifications = model.predict(val_images)
print(classifications[0])
print(test_labels[0])
```

→ Now convert it to the actual digit!

Issues with classification



Issues with classification

Steps to take:

1. Get as many examples of shoes as possible
2. Train using these examples
3. Profit!



Issues with classification

Steps to take:

1. Get as many examples of shoes as possible
2. Train using these examples
3. Profit!

```
Training accuracy: .920
Training accuracy: .935
Training accuracy: .947
Training accuracy: .961
Training accuracy: .977
Training accuracy: .995
Training accuracy: 1.00
```

Issues with classification

Steps to take:

1. Get as many examples of shoes as possible
2. Train using these examples
3. Profit!



```
Training accuracy: .920
Training accuracy: .935
Training accuracy: .947
Training accuracy: .961
Training accuracy: .977
Training accuracy: .995
Training accuracy: 1.00
```


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Training, validation & test data

The network 'sees' everything. Has no context for measuring how well it does with data it has never previously been exposed to.

Data

49

Training, validation & test data

The network 'sees' a subset of your data. You can use the rest to measure its performance against previously unseen data.



50

Training, validation & test data

The network 'sees' a subset of your data. You can use an unseen subset to measure its accuracy while training (validation), and then another subset to measure its accuracy after it's finished training (test).



Practice!
Add validation_data=(val_images, val_labels) as an argument to the fit function!

51

Some questions

Regression outputs a set of probabilities that predict the likelihood that the inputs belong to each possible class or label.

a) True

b) False

52

Some questions

Neural networks must have one input and one output.

a) True

b) False

54

Some questions

When should you stop training a model?

a) For as long as possible
b) For a particular number of epochs
c) Until the training error steps getting smaller
d) Until the validation error stops getting smaller

56

Some questions

The test data set is used to evaluate the model after each epoch.

a) True
b) False

58

Some questions

An overfit model will perform poorly on which of the following? Select all that apply.

a) Training data
b) Validation data
c) Test data
d) Data encountered in the real world after deployment

60

Some questions

Inference is when a model is used to make a prediction with some given input data.

a) True
b) False

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Challenges for TinyML

Microprocessor vs Microcontroller

Microprocessor

- Heart of a **computer system**
- Just the processor, memory and storage are **external**
- Mainly used in **general purpose systems** like laptops, desktops and servers
- **Offers flexibility** in design
- System size is **big**

Microcontroller

- Heart of an **embedded system**
- Memory and storage are all **internal** to the system
- Mainly used in **specialized, fixed function systems** like phones, MP3 players, etc.
- **Limited flexibility** in design
- **System size is tiny**

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Challenges for TinyML

Microprocessor vs Microcontroller

	Microprocessor	>	Microcontroller
Platform			
Compute	1GHz–4GHz	~10X	1MHz–400MHz
Memory	512MB–64GB	~10000X	2KB–512KB
Storage	64GB–4TB	~100000X	32KB–2MB
Power	30W–100W	~1000X	150µW–23.5mW

65

Challenges for TinyML

Microprocessor vs Microcontroller

Implications

- How complicated is the running task?
- How much memory does it need to have?
- How long does the job have to perform?

Microcontroller

1MHz-400MHz
2KB - 512KB
32KB - 2MB
150µW–23.5mW

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Challenges for TinyML

Software and Operating Systems

Mobile OS

Embedded Sys.

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Challenges for TinyML

Libraries

Portability Opportunity

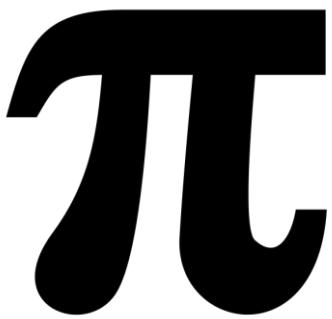
Able to execute the same code on different microprocessor hardware and architectures.

Challenges for TinyML

Number representation and calculations



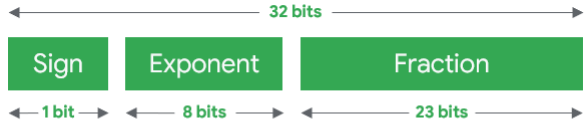
Single Precision
IEEE 754 Floating-Point Standard



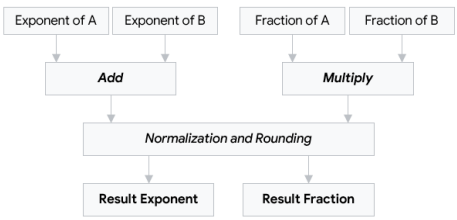


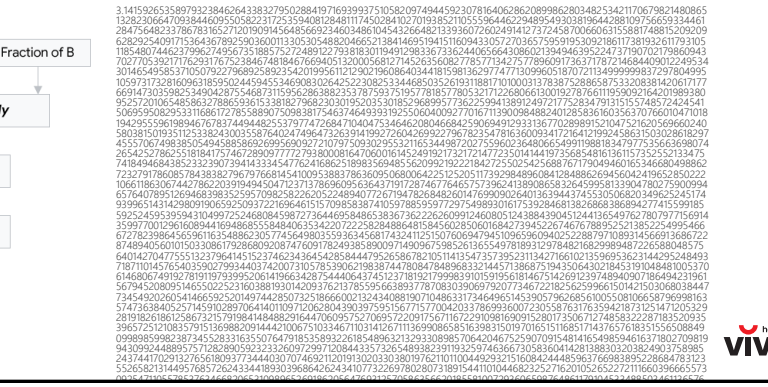
Challenges for TinyML

Number representation and calculations



Single Precision
IEEE 754 Floating-Point Standard





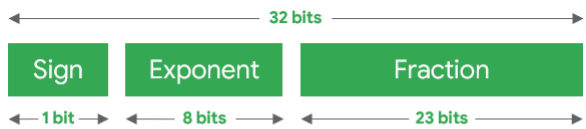
Getting started with DL

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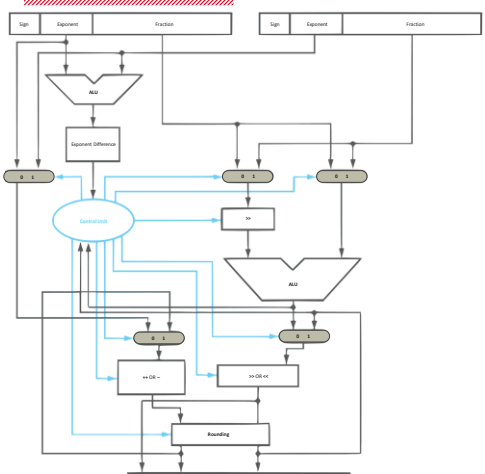
70

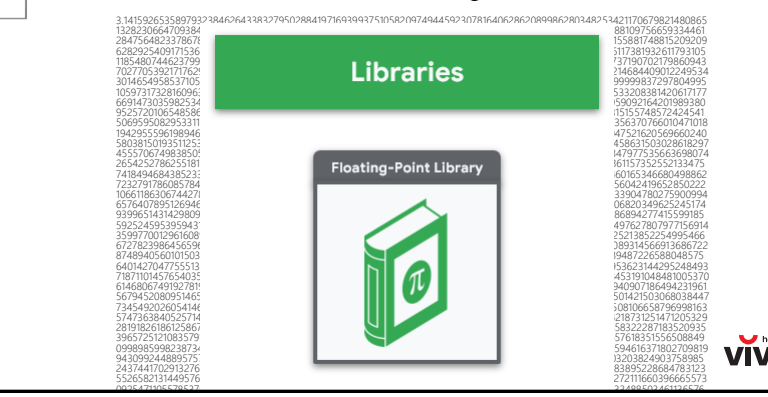
Challenges for TinyML

Number representation and calculations



Single Precision
IEEE 754 Floating-Point Standard





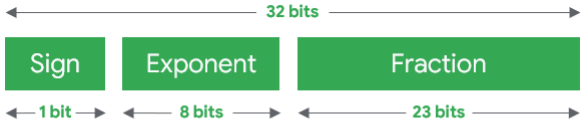
Getting started with DL

hogeschool vives

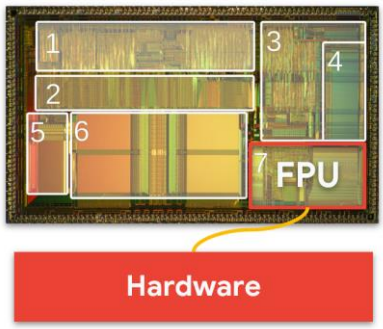
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Challenges for TinyML

Number representation and calculations



Single Precision
IEEE 754 Floating-Point Standard



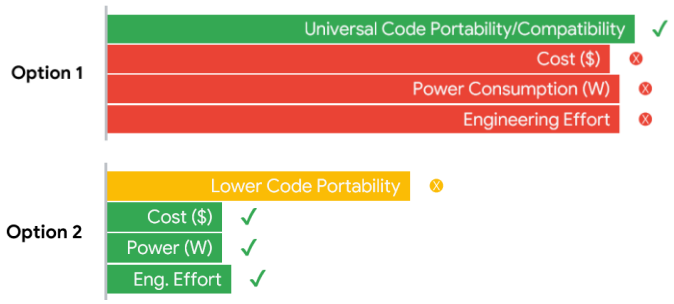


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Challenges for TinyML

Portability Trade-off(s)

Sacrifice portability across systems for efficiency in system performance and power efficiency

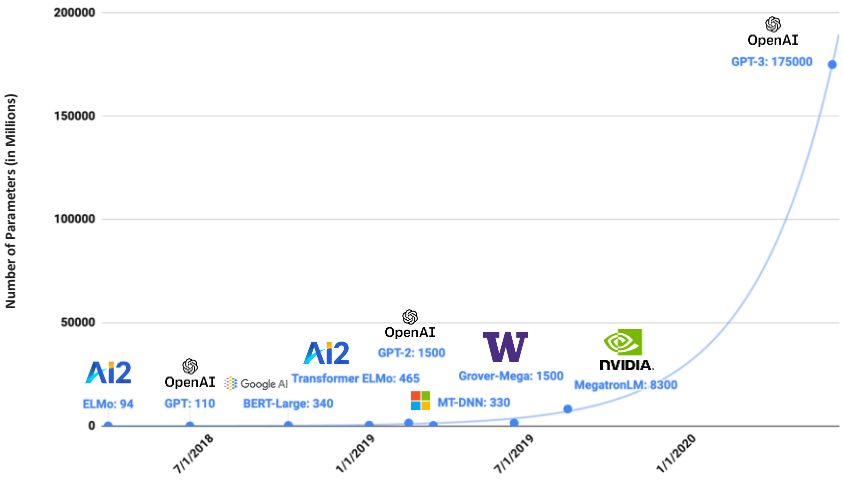


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Challenges for TinyML

Model sizes

ML Model Size Growth

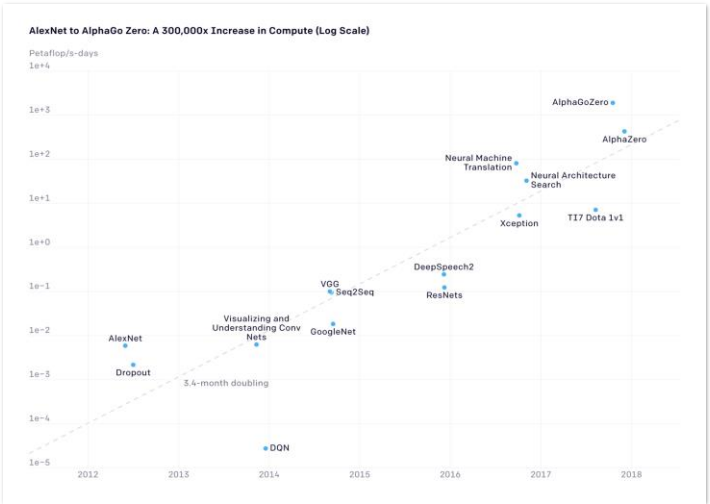


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Challenges for TinyML

Model sizes

ML Compute Needs (2012 to Present Day)



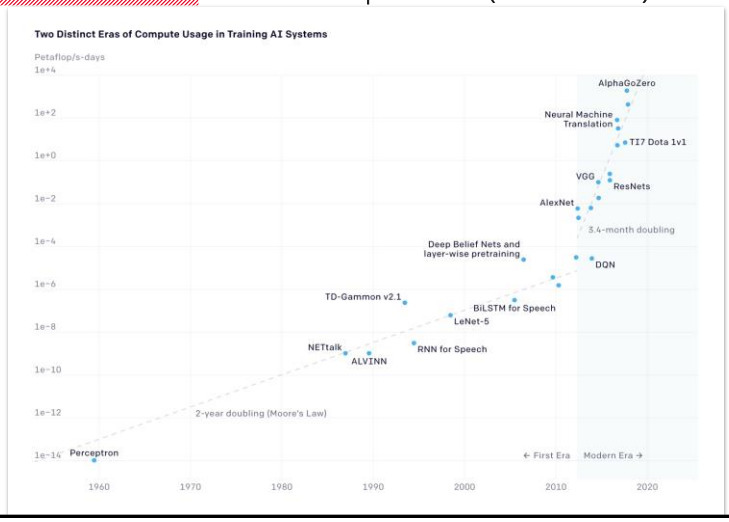
In recent years, computing required growing 300,000x to train the machine learning models that are widely deployed in the industry

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Challenges for TinyML

Model sizes

ML Compute Needs (from the 1960s)



In recent years, the amount of computing needed has grown remarkably fast.

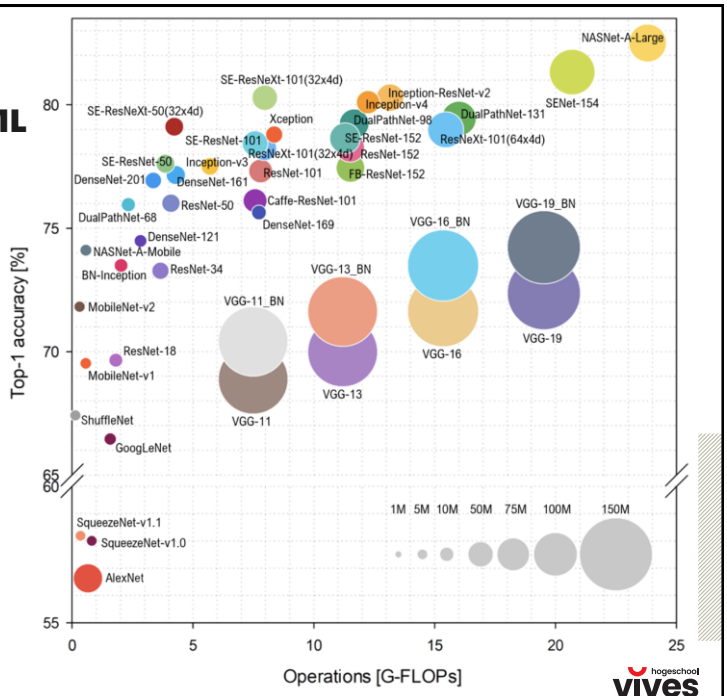
Compute requirements are doubling nearly every 3 to 4 months

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Challenges for TinyML

ML Model Evolution

- AlexNet (2012)
  - 57.1% accuracy
  - 61MB in size

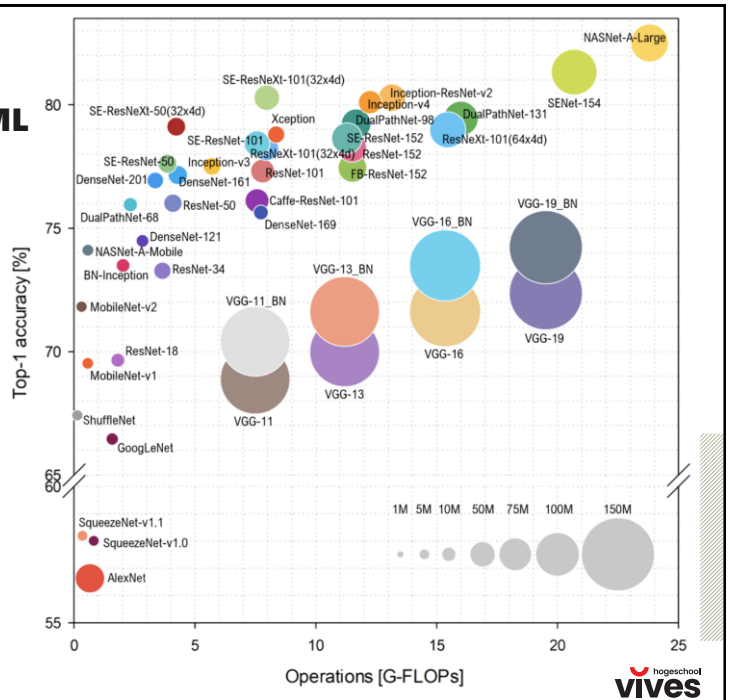


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Challenges for TinyML

ML Model Evolution

- ResNet (2015)
  - 75.8% accuracy
  - 22.7MB in size

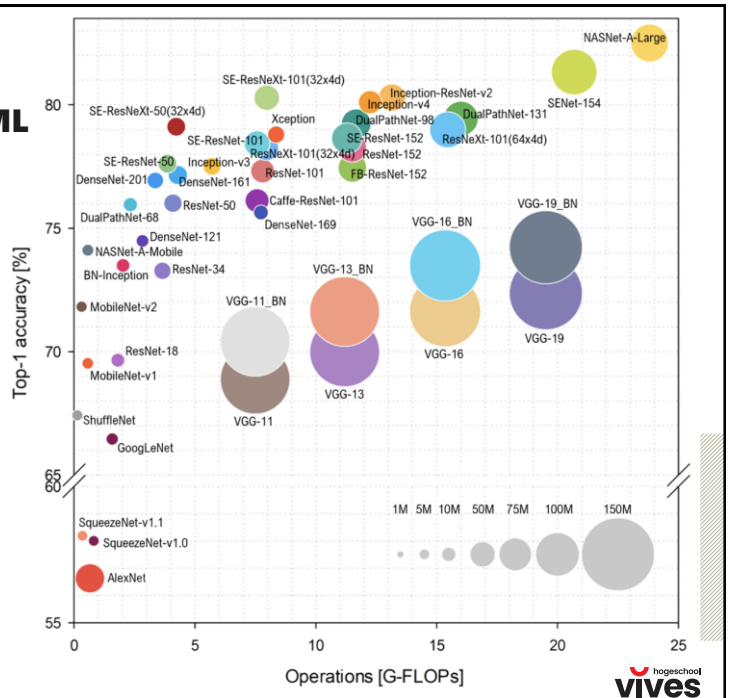


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Challenges for TinyML

ML Model Evolution

- MobileNet (2015)
  - *MobileNetv1*
    - 70.6% accuracy
    - 16.9MB in size



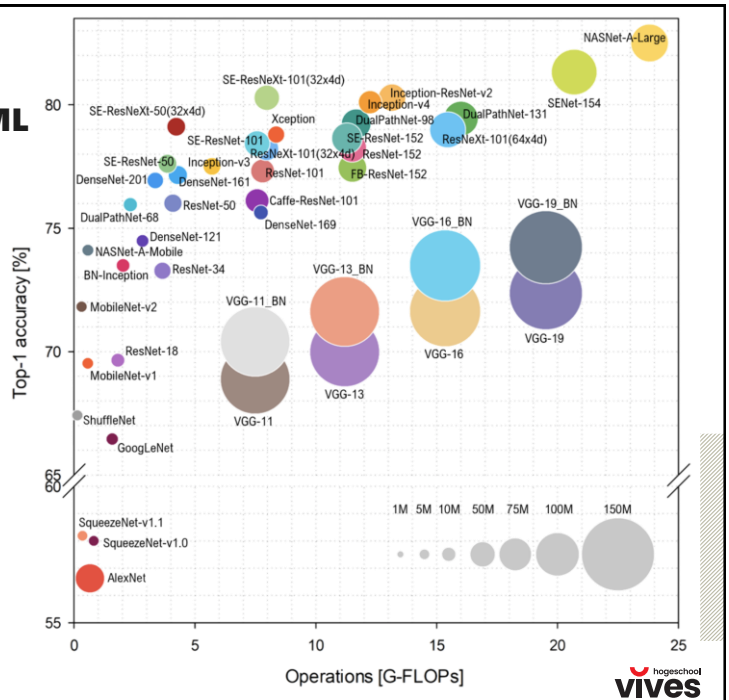
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Challenges for TinyML

ML Model Evolution

Do you see the problem?

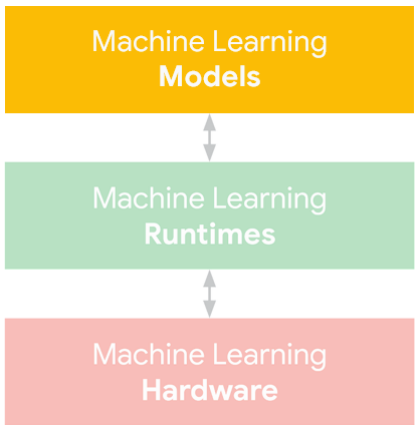
Our boards have only **256KB** of RAM (memory), yet MobileNetv1 needs **16.9MB**!



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Challenges for TinyML

How do we solve these problems?



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Challenges for TinyML

How do we solve these problems?

Model compression techniques

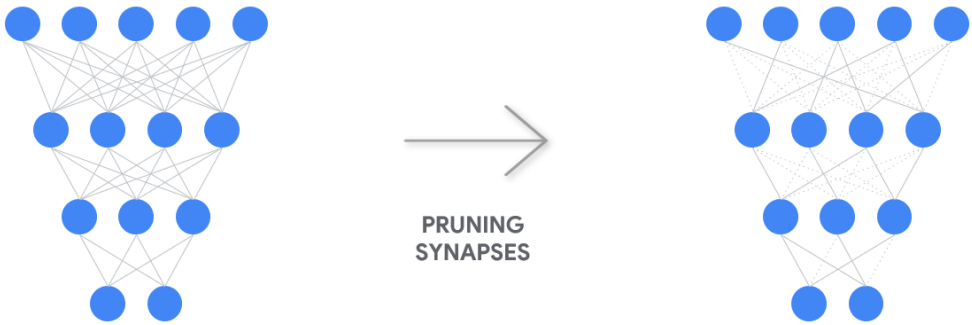
1. Pruning
2. Quantisation
3. Knowledge distillation
4. …

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Challenges for TinyML

How do we solve these problems?

1. Pruning

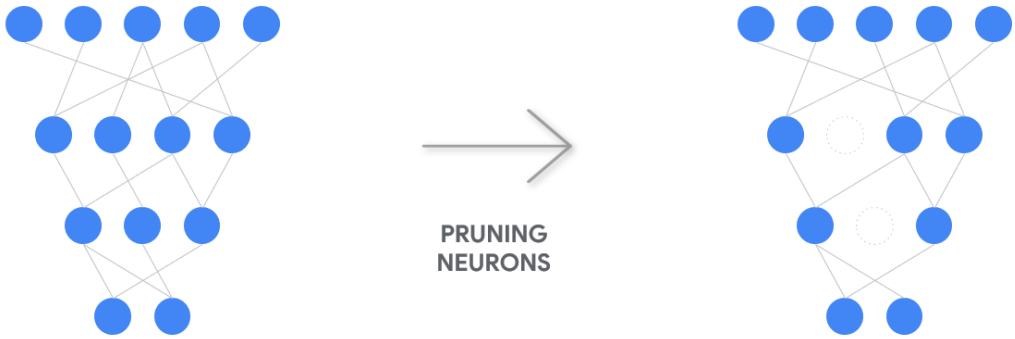


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Challenges for TinyML

How do we solve these problems?

1. Pruning

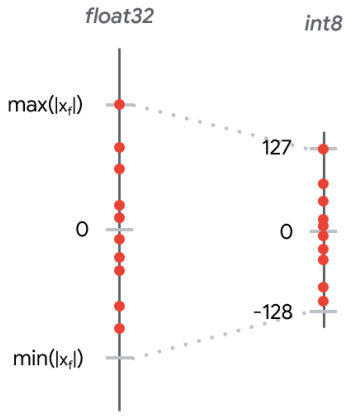


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Challenges for TinyML

How do we solve these problems?

2. Quantization

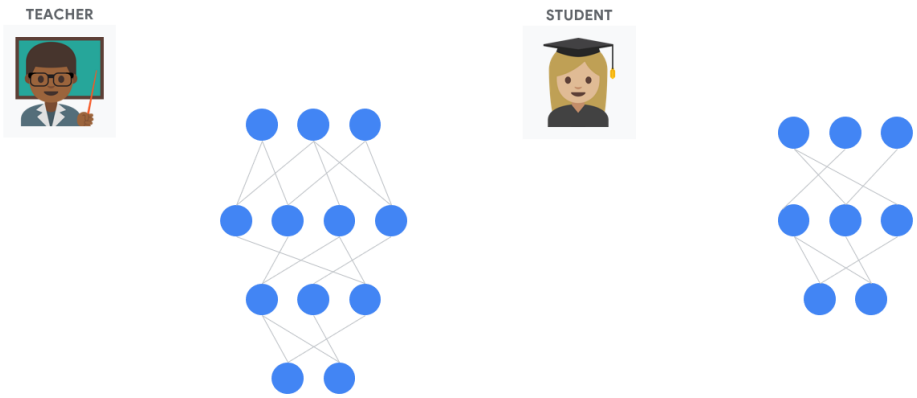


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Challenges for TinyML

How do we solve these problems?

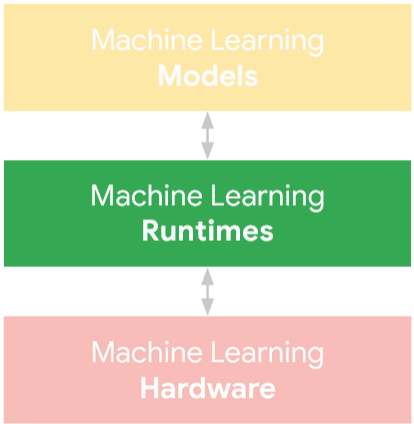
3. Knowledge Destillation



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Challenges for TinyML

How do we solve these problems?



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Challenges for TinyML

Runtimes

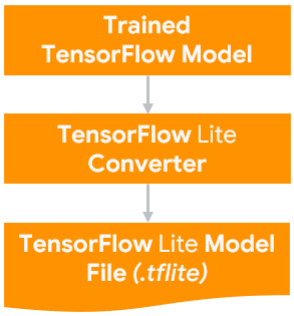
	TensorFlow	TensorFlow Lite
Topology	Variable	Fixed
Weights	Variable	Fixed
Binary Size	Unimportant	High Priority
Distributed Compute	Needed	Not Needed
Developer Background	ML Researcher	Application Developer

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Challenges for TinyML

Runtimes

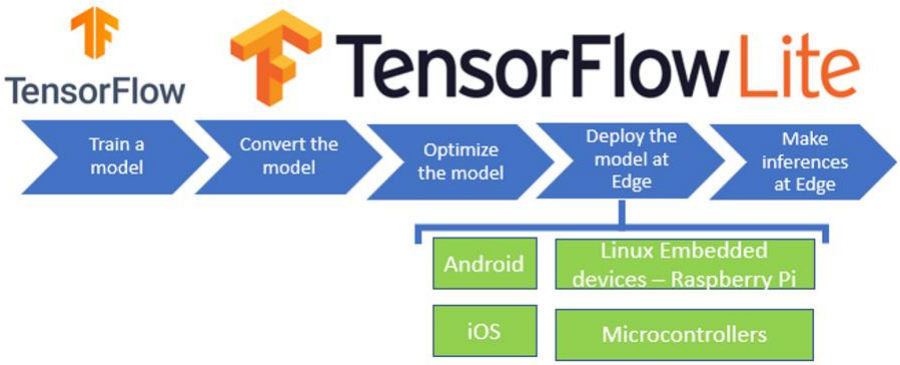
Architecture



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Challenges for TinyML

Runtimes



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Some questions

Machine learning models can usually be easily run on microcontrollers without any modification.

a) True

b) False

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Some questions

The most popular operating system for a microcontroller is:

a) Mac
b) Linux
c) Windows
d) None

93

Some questions

Embedded machine learning is limited to just accelerometer, audio, and image data.

a) True
b) False

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